

Administrator Manual — Tracker Porti

🇮🇹 Manuale in italiano: [manuale_admin.md](#) · [index.html](#)

Guide to the functions reserved for Tracker Porti **administrators**: user and group management, coverage-map controls, risk model, system logs, and editing configuration files.

This manual **complements** the [User manual](#), which covers day-to-day use of the app. Here you'll find **only** what is reserved for administrators. For common features (monitoring, followed ships, ship detail, notifications, export...) refer to the user manual.

Administrator role

Amministrazione utenti

← App · Esci

NOME	EMAIL / USERNAME	RUOLO	STATO	ONLINE	GRUPPO	AREE	AZIONI
Amministratore (tu)	admin@local admin	admin	active	●	—	1	Disabilita Rendi utente Reset password
Manuale Utente	manuale.utente@tracker-porti.local manuale_utente	user	active	●	Monitoraggio Toscana	1	Disabilita Rendi admin Reset password Impersona Elimina
Squadra Porto	squadra.porto@tracker-porti.local squadra_porto	user	active	●	Monitoraggio Toscana	1	Disabilita Rendi admin Reset password Impersona Elimina

Gruppi di utenti

I membri di un gruppo condividono (unione) aree di monitoraggio, navi seguite, navi flaggate, navi silenziate e le preferenze di notifica/mappa. Le impostazioni gestite dall'amministratore (sorgenti dati, pesi di rischio) restano globali. Un gruppo richiede almeno 2 utenti.

+ Nuovo gruppo

NOME	DESCRIZIONE	MEMBRI	AZIONI
		Manuale Utente	Squadra Porto

Administration page: user table with role, status, group, areas, and available actions.

Administrators see the **Admin** link at the top right, which opens the **administration page** (/admin). An administrator can:

- **approve** pending new registrations;

- **enable or disable** an account;
- **change a user's role** (regular user ↔ administrator ↔ tester);
- **reset a user's password** — generates a **one-time link** (valid 24 hours) to hand to them;
- **delete** a user;
- **create and manage user groups** (see below);
- **impersonate** a user — view their areas, monitoring, and followed ships in **read-only** mode, with a prominent banner and one-click exit;
- consult the **logs** (activity log and API log), shared and visible only to administrators.

Actions on a user row

Each table row shows **at most one highlighted button** — the one that row is waiting for: **Approve** for a pending registration, **Re-enable** for a disabled account. Everything else lives in the ... menu at the end of the row, always in the same order:

1. **status** — Approve / Approve as tester / Disable / Re-enable;
2. **role** — Make administrator, Revoke administrator, or **Promote to user** (for a tester account);
3. **utilities** — Reset password, Impersonate;
4. **destructive action** — Delete user (in red, with confirmation).

From tester to regular user

The **tester** role is only assigned at approval time, with **Approve as tester**: it is an account with reduced limits (number and size of areas, number of followed ships) meant for trials.

When you want to lift those limits, open the ... menu on the user's row and choose **Promote to user**: the account becomes a regular user, with no tester limits, keeping the areas, followed ships and settings already configured. The move can only be reversed through a new approval, so the tester role cannot be re-assigned to an account that has been promoted.

Settings managed by administrators (data sources, sanctions/PSC screening, risk-score weights) are **global** for all users. In the Settings interface, the tabs and toggles reserved for administrators are **hidden** from regular users (and protected server-side regardless).

User groups

Amministrazione utenti

← App · Esci

Amministratore (tu)

admin@local
admin

admin

active

●

—

1

Rendi utente

Reset password

Manuale Utente

manuale.utente@tracker-porti.local
manuale_utente

user

active

●

Monitoraggio Toscana

1

Disabilita

Rendi admin

Reset password

Impersona

Elimina

Squadra Porto

squadra.porto@tracker-porti.local
squadra_porto

user

active

●

Monitoraggio Toscana

1

Disabilita

Rendi admin

Reset password

Impersona

Elimina

Gruppi di utenti

I membri di un gruppo condividono (unione) aree di monitoraggio, navi seguite, navi flaggate, navi silenziate e le preferenze di notifica/mappa. Le impostazioni gestite dall'amministratore (sorgenti dati, pesi di rischio) restano globali. Un gruppo richiede almeno 2 utenti.

+ Nuovo gruppo

NOME	DESCRIZIONE	MEMBRI	AZIONI
Monitoraggio Toscana	Squadra che segue il traffico costiero toscano	Manuale Utente Squadra Porto + aggiungi membro... ▼	Rinomina Sciogli

Administration page: User groups section with the creation form and the list of existing groups.

An administrator can put several users into a **group**. When a user is part of a group, they **share with the other members** (as a **union** of what each one had):

- monitoring **areas**;
- **followed ships**, **flagged ships** ★, and **muted ships** 🚫 ;
- **notification preferences** (app and Telegram) and **map display**, plus the **default area**.

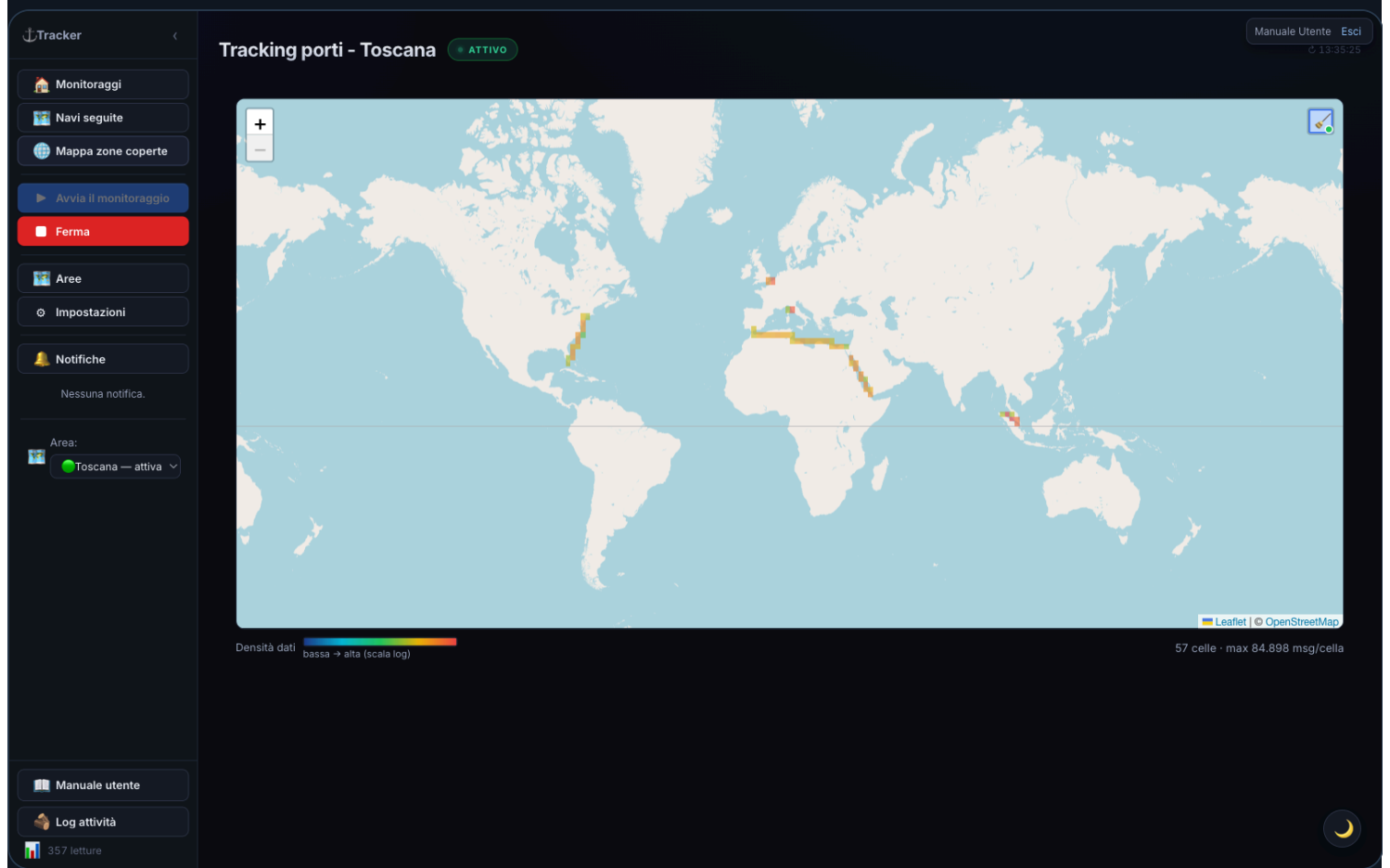
In practice: if one member adds an area, follows a ship, or enables a notification, **the other members get it** on their next visit — and vice versa (write-through sync). What stays **personal** to each user: the **Telegram link** (their own chat) and the interface **language**.

Management (from the Admin page): a group has a **name**, a **description**, and **at least 2 members**. On creation you choose the **template user** whose settings the group starts from. You can **add/remove** members or **dissolve** the group.

A removal that would leave **only 1 member** is blocked: in that case, **dissolve** the group instead. If an administrator removes a user from a group, that user **keeps** everything that was shared up to that point (areas, ships, settings): they simply stop syncing with the others.

Group activity log: every write-through share is also recorded (who, when, what) — members review it themselves in the Group activity section (visible to anyone in a group, no admin action needed). Rows older than `GROUP_ACTIVITY_LOG_RETENTION_DAYS` days (default 90, in `app.config.properties`) are deleted automatically.

Coverage map — administrator controls



World coverage map: grid colored by AIS message density.

The Coverage map is visible read-only to all users (and even without login at `/heatmap`). Only **administrators** can additionally:

- **start** and **stop** data collection (**Start/Stop collection** buttons);
- see **real-time connection statistics** (bandwidth used, messages per second, populated cells...);
- **Refresh map** and **Clear data** (empties the collected grid).

How collection works: once started, it runs **in the background** on the server — even if no one has the page open — until an administrator stops it. It resumes on its own after a server restart. As a safety measure, it **automatically shuts off** if no user has been active for **10 minutes**.

Warning: while collection is active, the app **downloads data from the whole world** continuously (roughly **200–400 MB per hour**). The feature requires an **AISStream key from a separate account** (different from the one used for regular monitoring), set in `local.properties` as `HEATMAP_AIS_API_KEY`. Without that key, the feature stays off.

Port discovery (per area)

The system figures out on its own, in the background, which real ports fall inside each monitored area. It's not just internal data: **fallback mode** (below) uses it to discover ships AIS has never seen, when an area hasn't had signal for a while — on the **Areas** screen a  icon next to that area's fallback toggle warns you if its port isn't (yet) resolved on MyShipTracking, meaning that part of the feature isn't active there (repositioning already-known ships still works).

How ports are identified: if the area already has berths really observed by AIS, those (grouped by proximity) **are** the ports — no external source is contacted automatically. Otherwise it cross-checks World Port Index, UN/LOCODE and VesselFinder (Global Fishing Watch is a candidate in the code but **isn't reachable today** — its anchorages dataset isn't available via API); a point found by at least two independent sources is considered confirmed, one found by a single source stays "needs review".

When automatic discovery runs:

- **When a new area is created.**
- **In the background at server restart**, one time only, for existing areas that don't have discovered ports yet (one area at a time, with a pause between each, so it doesn't slow down startup).

Ports panel (Areas screen)

Tracker

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Log attività

7344 letture

AmministratoreAdminEscl

Porti — Toscana

Porti identificati per quest'area (services/port-discovery.js). Evidenziati sulla mappa qui sopra, colorati per stato.

Cerca porti ora

Confermare/rifiutare decide se questo porto viene usato per scoprire navi mai viste dall'AIS (arrivi/partenze) quando l'area è in fallback — non tocca il riposizionamento delle navi già note, che non dipende dai porti. La decisione è reversibile in qualsiasi momento.

NOME	COORDINATE	STATO	SORGENTI
42.11, 11.76	42.1189, 11.7633	CONFERMATO	berths
42.59, 10.10	42.5892, 10.0988	CONFERMATO	berths
42.73, 10.16	42.7291, 10.1578	CONFERMATO	berths
42.74, 10.24	42.7447, 10.2426	CONFERMATO	berths
42.81, 10.32	42.8124, 10.3239	CONFERMATO	berths
42.92, 10.25	42.9226, 10.2526	CONFERMATO	berths
Test Port	43.5000, 10.3000	RIFIUTATO	berths, gfw

Confronta con sorgenti esterne

Cerca porti per quest'area su LOCODE/WPI/Global Fishing Watch/VesselFinder, indipendentemente dalle banchine già rilevate dall'AIS, e mostrali qui accanto (in blu tratteggiato sulla mappa) per confrontarli. Non toccano l'elenco sopra finché non li aggiungi esplicitamente.

Cerca in sorgenti esterne ora

NOME	SW (LAT, LON)	NE (LAT, LON)	PAROLA CHIAVE	STATO	FALLBACK	DATI SALVATI	AZIONI	
Liguria (test)	44.2737, 8.5625	44.5156, 9.3858	—	spenta	In fallback (forzata)	nessun porto	272 letture · 5 navi	
Toscana (in uso)	42.0860, 9.8108	43.0508, 12.3459	—	attiva	AIS attivo	porto da risolvere	7.1k letture · 273 navi	

Areas — Ports panel: confirmed/rejected list and compare against external sources, with markers on the map.

- From the **Areas** screen, the  button on each row opens a panel dedicated to that area's ports, below the map (not a popup that hides it — the map stays visible with its markers highlighted above the table):
- Port list** — name, coordinates, status (**Confirmed** green / **Needs review** amber / **Rejected** red) and the sources that found it. Confirming or rejecting decides whether that port is queried to discover ships AIS never reported while the area is in fallback — it doesn't affect repositioning already-known ships, which doesn't depend on ports. The decision is reversible at any time (a rejected port can be confirmed again later).
 - Search ports now** — re-runs the automatic discovery described above (same logic: berths if already observed, otherwise the external cascade). If the area had accumulated duplicate ports from past runs (the same spot saved multiple times under different names), this action cleans them up automatically, replacing them with the current re-clustered set — as long as they haven't already been manually confirmed/rejected.
 - Compare against external sources** — a separate section, always available (even when the area already has observed berths, unlike the automatic discovery above): the **Search external sources now** button queries LOCODE/WPI/Global Fishing Watch/VesselFinder independently and shows results **live** as they arrive (LOCODE/WPI/GFW almost instant, VesselFinder slower due to its anti-ban pacing). Each candidate has a checkbox and an **Add** button; **Add selected** adds them in a batch. An added candidate goes straight

to **Confirmed** — the admin's pick already is the review. Candidates don't touch the list above until you explicitly add them.

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7344 letture

Confronta con sorgenti esterne

AmministratoreAdminEsci

Cerca porti per quest'area su LOCODE/WPI/Global Fishing Watch/VesselFinder, indipendentemente dalle banchine già rilevate dall'AIS, e mostrali qui accanto (in blu tratteggiato sulla mappa) per confrontarli. Non toccano l'elenco sopra finché non li aggiungi esplicitamente.

22 trovati finora...

Ricerca in corso...

Aggiungi selezionati

Capraia 43.8500, 9.8333	LOCODE	Aggiungi
Castiglione della Pescaia 42.7667, 10.8833	LOCODE	Aggiungi
Cavo 42.8500, 10.4167	LOCODE	Aggiungi
Civitavecchia 42.1000, 11.8000	LOCODE	Aggiungi
Follonica 42.9167, 10.7500	LOCODE	Aggiungi
Isola del Giglio 42.3667, 10.9000	LOCODE	Aggiungi
Marciana Marina 42.8000, 10.2000	LOCODE	Aggiungi
Marina di Campo 42.7333, 10.2333	LOCODE	Aggiungi
Marina di Grosseto 42.7167, 10.9833	LOCODE	Aggiungi
Montalto di Castro 42.3500, 11.6167	LOCODE	Aggiungi
Orbetello 42.4500, 11.2167	LOCODE	Aggiungi
Pianosa 42.5667, 10.0667	LOCODE	Aggiungi
Porto Ercole 42.3833, 11.2000	LOCODE	Aggiungi
Vetralia 42.3167, 12.0500	LOCODE	Aggiungi
Civitavecchia 42.1020, 11.7710	WPI	Aggiungi
Giglio Porto 42.3620, 10.9210	WPI	Aggiungi
Portoferraio 42.8167, 10.3333	WPI	Aggiungi
Porto Santo Stefano 42.4400, 11.1220	WPI	Aggiungi

Areas — compare against external sources: candidates arriving live with checkboxes, single or batch add.

On the Areas screen's map, while the panel is open, already-confirmed ports show as solid circles colored by status; freshly-found external candidates show as **dashed blue** circles, so you can visually compare before deciding whether to add them.

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AGGIUNGI AREA

Inserisci le coordinate in gradi decimali, oppure inquadra l'area sulla mappa e premi "Cattura vista corrente". Tocca una riga dell'elenco per caricarla qui e modificarla.

Nome area
Es. Porto di Ancona

Parola chiave (facoltativa)
Es. ANCONA

SW Lat
40.95

SW Lon
16.60

NE Lat
41.30

NE Lon
17.10

Cattura vista corrente

+ Aggiungi area

+

-

Parc naturel marin du Cap Corse et de l'Agriote

Parco Nazionale dell'Arcipelago Toscano

Corse

Parc naturel régional de Corse

Porto di Ancona

Porto di Genova

Porto di Livorno

Porto di Pisa

Porto di Roma

Porto di Venezia

Amministratore Admin Esci

Porti — Toscana

Porti identificati per quest'area (services/port-discovery.js). Evidenziati sulla mappa qui sopra, colorati per stato.

Cerca porti ora

Confermare/rifiutare decide se questo porto viene usato per scoprire navi mai viste dall'AIS (arrivi/partenze) quando l'area è in fallback — non tocca il riposizionamento delle navi già note, che non dipende dai porti. La decisione è reversibile in qualsiasi momento.

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42.59, 10.10	42.5892, 10.0988	• CONFERMATO	berths
42.73, 10.16	42.7291, 10.1578	• CONFERMATO	berths
42.74, 10.24	42.7447, 10.2426	• CONFERMATO	berths

Areas — map with confirmed ports (solid) and external candidates (dashed blue) side by side.

Risk model (signal weights)

Besides the **cargo-type weights** (editable by everyone in Settings), administrators can adjust **how many points each risk signal is worth** (AIS blackout, spoofing/position jump, dwell, draught increase, sanctions, PSC, GFW events, rendezvous, etc.) from ⚙ **Settings** → “⚓ Risk model” section.

- A **grid** with one field per signal: change the values and press  **Save weights**. **Immediate** effect, no restart needed. **Reset to default** restores factory values.
- Setting a weight to **0** disables that signal.
- You can save full configurations as **risk profiles** (*Risk profile* menu → *Save as...*) and recall them with *Apply* — useful for quickly switching between different setups (e.g. a more aggressive profile, or one tuned for areas with poor AIS coverage).

Detection thresholds and signal **multipliers** are not in the UI: they stay in the `app.config.properties` file (`RISK_*` keys).

The Settings **General** tab, “⚠️ Risk model” section, also has two toggles **reserved for administrators** (invisible to other users):

- **Exclude tankers** — doesn’t assign the “ship type” score to tanker hulls (AIS code 80–89). Useful when monitoring weapons transport, which a tanker can’t carry out: it only zeroes that one factor, not the whole score (a tanker can still land in the red band from other signals — sanctions, AIS dark, etc.). It’s a **global** value shared by every user, and different from the notifications’ ship-type filter (Settings → **Notifications**, personal to each user): that one only decides what reaches you as a notification, without touching the score.
- **Check position jump / Check AIS blackout** — include those signals in the score. Disable them in areas with poor AIS coverage, where sparse reports produce false positives (apparent jumps/gaps that aren’t real).

System logs

Logs are **shared** and visible **only to administrators**, as tabs in Settings.

Activity log

Tracker

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GNV ORION spostata da — a toscana

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Manuale utente

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871 letture

Tracking porti - Toscana

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Log attività

Registra le operazioni significative dell'app (stream, recupero dati, sanzioni, backup, errori) in un file con rotazione automatica (max ~5 MB). Attivo per impostazione predefinita.

LIVE

Svuota log

14:33:20 28/07/26 SCRAPE ShipFinder ok per STELLA MARIS {"mmsi":247152290,"imo":null}

14:33:30 28/07/26 PORTO 1 arrivi: STELLA MARIS {"navi":1}

14:33:31 28/07/26 SCRAPE VesselFinder fallito per STELLA MARIS: Timeout {"mmsi":247152290}

14:33:35 28/07/26 SCRAPE Global Fishing Watch ok per STELLA MARIS {"mmsi":247152290,"imo":null}

14:34:30 28/07/26 BERTHS Ricalcolo incrementale banchine per arrivi/partenze {"aree":["toscana"]}

14:36:30 28/07/26 PORTO 1 arrivi: COLLECT WAVES {"navi":1}

14:36:30 28/07/26 BERTHS Ricalcolo incrementale banchine per arrivi/partenze {"aree":["toscana"]}

14:38:30 28/07/26 PORTO 1 arrivi: MOBY NIKI {"navi":1}

14:38:30 28/07/26 BERTHS Ricalcolo incrementale banchine per arrivi/partenze {"aree":["toscana"]}

14:46:43 28/07/26 SCRAPE MyShipTracking ok per EUROCARGO GENOVA {"mmsi":247293800,"imo":null}

14:46:44 28/07/26 SCRAPE ShipFinder ok per EUROCARGO GENOVA {"mmsi":247293800,"imo":null}

14:46:44 28/07/26 SCRAPE MarineTraffic ok per EUROCARGO GENOVA {"mmsi":247293800,"imo":null}

14:46:49 28/07/26 SCRAPE Global Fishing Watch ok per EUROCARGO GENOVA {"mmsi":247293800,"imo":null}

14:46:54 28/07/26 SCRAPE VesselFinder fallito per EUROCARGO GENOVA: Timeout {"mmsi":247293800}

14:47:30 28/07/26 PORTO 1 arrivi: EUROCARGO GENOVA {"navi":1}

14:48:27 28/07/26 AUTH Nuova registrazione in attesa di approvazione: squadra.porto@tracker-porti.local {"userId":3}

14:48:30 28/07/26 BERTHS Ricalcolo incrementale banchine per arrivi/partenze {"aree":["toscana"]}

14:49:10 28/07/26 AUTH Accesso: admin@local {"userId":1}

Settings — Activity log: chronological record of the app’s operations.

Records the app’s significant operations (streams, data fetching, sanctions, backups, errors) in a file with **automatic rotation** (max ~5 MB). On by default (**Activity log** toggle in the General tab, admin). The log can also be viewed from the 🪵 **Activity log** overlay in the sidebar and can be **cleared**.

The log also records the **area-change notifications discarded** by the filters (Area change not notified: ...) with the reason: overlapping areas , only crossed , call too old . That is where to look when a user reports a missing alert they expected — see the AREA_CHANGE_* parameters in [app.config.properties](#) .

API log

Tracker

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Manuale utente

Log attività

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Cancellog

#	ORA	METODO	PATH	STATUS	DURATA
29641	14:50:49 28/07/26	GET	/api/stream/status?lang=it	200 OK	0ms
29642	14:50:49 28/07/26	GET	/api/areas?lang=it	200 OK	2ms
29643	14:50:49 28/07/26	GET	/api/ships/past/count?area=toscana&lang=it	304 Not Modified	1ms
29644	14:50:54 28/07/26	DB	/ais/toscana/PositionReport	200 OK	1ms
29645	14:50:57 28/07/26	DB	/ais/toscana/ShipStaticData	200 OK	1ms
29646	14:51:05 28/07/26	GET	/api/app-log?limit=1000&lang=it	200 OK	44ms
29647	14:51:07 28/07/26	DB	/ais/toscana/PositionReport	200 OK	3ms
29648	14:51:10 28/07/26	DB	/ais/toscana/PositionReport	200 OK	0ms
29649	14:51:24 28/07/26	DB	/ais/toscana/StandardClassBPositionReport	200 OK	1ms
29651	14:51:31 28/07/26	AIS	/ais/monitoring/heartbeat	200 OK	0ms

Settings — API log: list of /api calls with method, path, and outcome.

Lists the application’s API calls in real time (method, path, status, time), useful for diagnostics. Sensitive request bodies (login, etc.) are **suppressed**: passwords are never logged.

AIS diagnostics

Tracker

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GNV ORION spostata da — a toscana 12:52:23 28/07/26

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Manuale amministratore

Manuale utente

Log attività

494 letture

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Log API

Diagnostica AIS

Monitoraggio aree (toscana)

Chiave: _cf80 (local.properties) - local.properties

CONNESSIONE

● Connesso 13:38:31 28/07/2026

FRAME WS RICEVUTI

120

VELOCITÀ MESSAGGI

7 msg/min

LETTURE TOTALI DB

506

UPTIME SESSIONE

17m 39s

MESSAGGI NAVE (SESSIONE)

120

TOTALE RICONNESSIONI

0

ULTIMO ERRORE AIS

Nessuno

Navi seguite

Chiave: _7cea (local.properties) - local.properties

CONNESSIONE

● Disconnesso

FRAME WS RICEVUTI

0

VELOCITÀ MESSAGGI

0 msg/min

NAVI SEGUITE

0

ULTIMO ERRORE AIS

Nessuno

UPTIME SESSIONE

11m 17s

MESSAGGI NAVE (SESSIONE)

0

TOTALE RICONNESSIONI

0

IN RI-ACQUISIZIONE

0

Mappa zone coperte

Chiave: _96de (local.properties) - local.properties

CONNESSIONE

● Disconnesso

UPTIME SESSIONE

0s

Settings — AIS Diagnostics tab: connection status, uptime, frames received, and reconnections.

The **Settings → AIS Diagnostics** tab (visible only to administrators) shows the active area's data stream connection status, refreshed every 5s:

- **Connection** — Connected / Disconnected
- **Session uptime** — how long the stream has been active
- **WS frames received / Ship messages / Message rate**
- **Reconnections** — how many times the connection was restored
- **Last error** — the most recent error, if any

The **AIS outage banner** shown to all users on monitoring pages when an area goes quiet for a long time is described from the user's perspective in the [user manual](#); the external-confirmation mechanism behind it is in the [Credits](#) section below.

Fallback mode (per area, silent)

Fallback mode is **not a single switch for the whole site**: every monitored area has its own, independent of the others. When an area hasn't received a real AIS signal for a while (configurable threshold, AREA_SILENT_THRESHOLD_MIN — see [Editing configuration files](#)) — or when it's been manually **forced** (below) — the app **adds** scraping for that area on top of the same AIS feed, which keeps being listened to without interruption:

- **relocates** already-known ships via scraping on **ShipFinder** and **MyShipTracking** (the only two external sources that provide coordinates);
- **discovers** ships AIS never saw at all, by querying the area's port arrivals/departures on MyShipTracking — this only works if the area has a recognized, resolved port (see [Port discovery](#) above; the $\triangle$ icon on the Areas screen warns you when it's missing).

This covers both temporary AIS outages and **zones with structurally poor or absent AISStream coverage** (no nearby receiver): such an area stays "in fallback" continuously, silently, without raising any alarm — that's the intended behavior, not a malfunction.

Tracker

Monitoraggi

Navi seguite

Mappa zone coperte

Avvia il monitoraggio

Ferma

Aree

Impostazioni

Modalità fallback

Notifiche

Area: Toscana — attive

Manuale amministr...

Manuale utente

Log modalità fallback

Log attività

7344 letture

Tracking porti - Toscana

AGGIUNGI AREA

Inserisci le coordinate in gradi decimali, oppure inquadra l'area sulla mappa e premi "Cattura vista corrente". Tocca una riga dell'elenco per caricarla qui e modificarla.

Nome area

Es. Porto di Ancona

Parola chiave (facoltativa)

Es. ANCONA

SW Lat

40.95

SW Lon

16.60

NE Lat

41.30

NE Lon

17.10

Cattura vista corrente

+ Aggiungi area

NOME	SW (LAT, LON)	NE (LAT, LON)	PAROLA CHIAVE	STATO	FALLBACK	DATI SALVATI	AZIONI
Liguria (test)	44.2737, 8.5625	44.5156, 9.3858	—	spenta	In fallback (forzata)	272 letture · 5 navi	
Toscana (in uso)	42.0860, 9.8108	43.0508, 12.3459	—	attiva	AIS attivo	71k letture · 273 navi	

Areas — Fallback column: on/off toggle, ⚡ force toggle, and a live status label with port badge.

Two independent toggles, on the Areas screen, one row per area (both administrator-only — a regular user manages their own areas but doesn't see either):

- **Main toggle (on by default)** — enables/disables fallback scraping for that area entirely. Off: this area never gets scraping, even during a long AIS outage (followed ships passing through it are still tracked as always, regardless of this toggle — they aren't tied to an area). AIS listening itself is **never** affected by this toggle.
- ⚡ **"force" toggle** (only shown while the main one is on, **off by default**) — treats the area as silent even while AIS is receiving messages normally. Useful for a spot you know has thin real AIS coverage (few transponder-equipped ships nearby), where the automatic silence threshold would never trigger on its own despite needing scraping. Here too, AIS keeps being listened to exactly as before — scraping is simply added on top, in parallel.

How to tell if an area is using fallback right now: in the **Fallback** column on the Areas screen, a live label per row — **AIS active** or **In fallback** (with a duration when the silence is genuine, or "(forced)" when the state comes from the ⚡ toggle) — plus an /△ badge if a resolved port is missing for discovering new ships. On the **Ships present** page, the title also shows the area's name and — administrators only — the same "📡 Data from AIS" / "🔍 Data from fallback (scraping)" label. Individual ships show a small badge when their table position comes from ShipFinder/MyShipTracking instead of AIS (the same indicator already used for followed ships).

Diagnostic panel — ⚙ **Settings** → **AIS Diagnostics** (administrators only):

Tracker

Monitoraggi

Navi seguite

Mappa zone coperte

Avvia il monitoraggio

Ferma

Aree

Impostazioni

Modalità fallback

Notifiche 99

Area: Toscana — attiva

Manuale amministratore...

Manuale utente

Log modalità fallback

Log attività

7344 letture

Scraping status 24h

Recupero dati dai fornitori esterni (non dal feed AIS). Conteggio per fonte nelle ultime 24h.

Source	Count
VESSELFINDER	9
SHIPFINDER	58
MARINETRAFFIC	9
MYSHIPTRACKING	62

Amministratore Admin Esci

Modalità fallback

Per ogni area in fallback (inerte AIS reale da un po', o forzata manualmente — vedi schermata Aree per i due interruttori): riposiziona e scopre navi ANCHE via scraping ShipFinder/MyShipTracking, in aggiunta all'AIS, che resta comunque sempre ascoltato.

Area	Status
TOSCANA	AIS attivo
LIGURIA (TEST)	In fallback

SHIPFINDER

nessun problema · nessun problema in questa sessione

Volume scraping reale (ultime 48h, per fonte)

Riuscite Fallite

ShipFinder

14:00 23:00

MyShipTracking

14:00 23:00

Stima richieste/ora (navi seguite + navi nelle aree silenziose)

Stima richieste/ora (navi seguite + navi nelle aree silenziose)

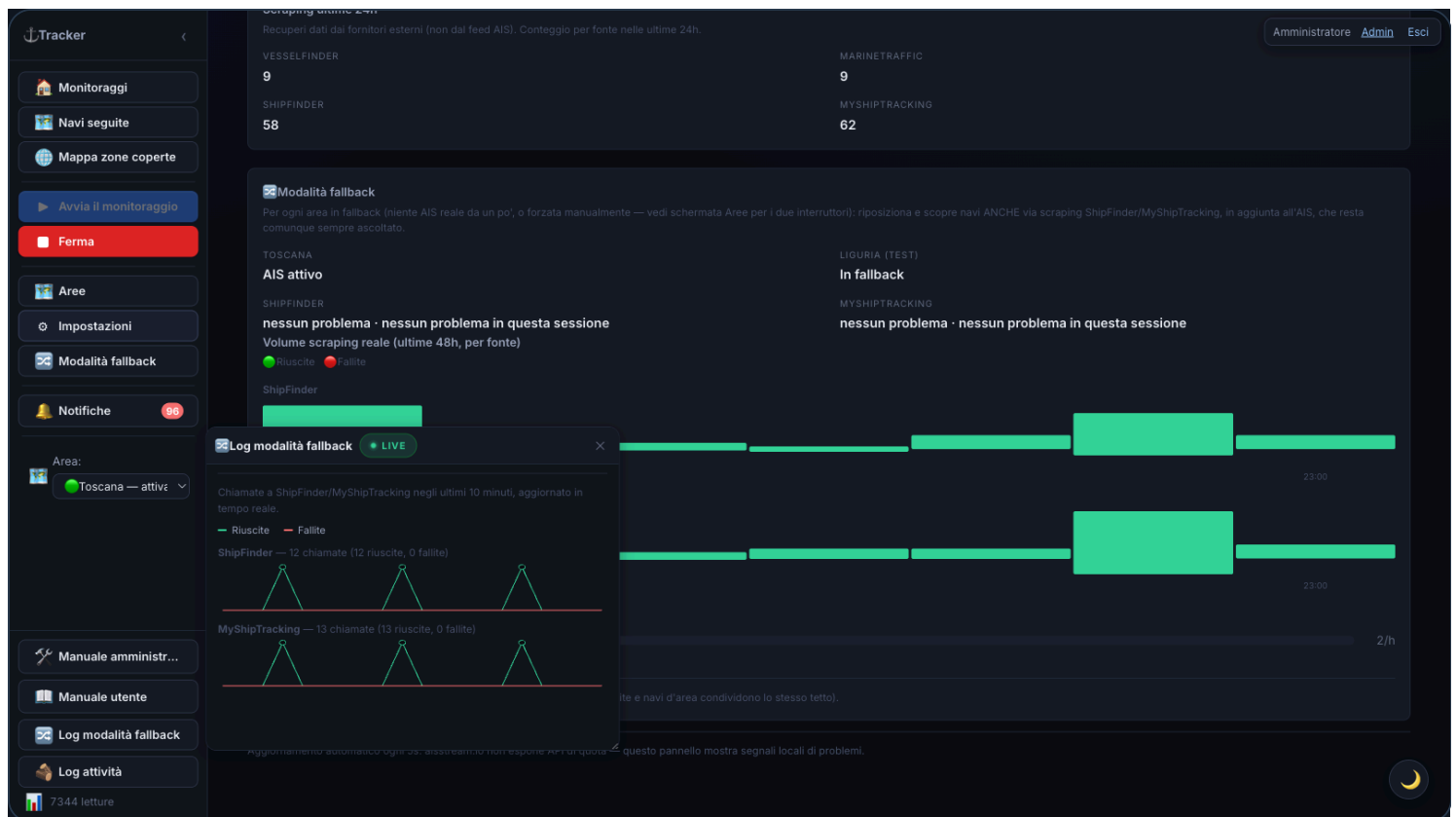
Tetto massimo configurato: 90 richieste/ora (SF+MST insieme, navi seguite e navi d'area condividono lo stesso tetto).

Aggiornamento automatico ogni 5s. aisstream.io non espone API di quota — questo pannello mostra segnali locali di problemi.

Settings → *AIS Diagnostics* — *Fallback mode section: per-area status, circuit breakers, scrape history, requests/hour estimate.*

- **Per-area status** — a list of every monitored area with, for each, the "AIS active"/"In fallback" label and since when (absent when the state comes from the ⚡ force toggle rather than genuine silence).
- **Real scrape history** — a separate bar chart for **ShipFinder** and **MyShipTracking**, last 48 hours, with stacked green/red bars for succeeded/failed requests and hour labels every 6 hours — hover a bar for the exact breakdown. Aggregated across all areas, not split per area.
- **Requests/hour estimate** — how many requests/hour followed ships plus ships in currently-silent areas add up to right now, against the configured hourly cap. More silent areas at once (by real threshold or by the force toggle) do **not** raise the hourly request cap (`FALLBACK_MAX_REQ_PER_HOUR`) — they only redistribute it over more ships, each then revisited less often.
- **Circuit-breaker status and session problems** — if ShipFinder or MyShipTracking show too many 403/429 errors in a short window (a possible sign of blocking), that source is automatically paused for a while, across all areas at once; here you see whether it's paused and until when, plus a **counter of how many suspected blocks happened this session** per source — it stays visible even after the block resolved itself, instead of disappearing the moment the source recovers.

Fallback mode log (real-time): a dedicated sidebar entry, above "Activity log" —  **Fallback mode log** (administrators only, always available, not just while some area is silent). It opens a floating window (draggable, like Activity log) with a small **line chart** — one for ShipFinder, one for MyShipTracking, with a green/red legend — that updates **in real time** on every actual call, whichever area or ship triggered it: succeeded/failed calls over the last 10 minutes, plus the session's total count per source. Desktop window, like Activity log it isn't designed for small screens.



The “Fallback mode log” window: a real-time line chart with a succeeded/failed legend.

The **Activity log** (see above) also records every successful reposition (Fallback: posizione ShipFinder/MyShipTracking per <ship>), with the **area** — **and, if identifiable within a few km, the nearest port** — in brackets whenever the ship belongs to an area (not for followed ships, which aren’t tied to one).

If a source is paused for suspected blocking, you get a dedicated alert — **administrators only** — via in-app notification, Telegram (toggle in Settings → External integrations → Telegram, “suspected ban” entry) and a hint in the AIS-outage banner while logged in as an admin.

i The **AIS outage banner** (shown to every user, when the whole AISStream service has a problem) remains a **separate**, unchanged mechanism — see above. Per-area fallback mode raises no notifications of its own: it’s designed to stay silent even while continuously active on a poorly-covered area.

Editing configuration files

Some advanced settings aren't in the interface but in text files in the project folder. Open them with a text editor, change the values, and **restart the application** to apply them. Lines starting with `#` (or `//`) are comments and are ignored.

`local.properties` — keys and secrets

Contains the API key and initial preferences. Format `KEY=value`, one per line. **Must not be shared** (it contains the API key) and is in `.gitignore`. If it doesn't exist, copy it from `local.properties.example`.

Key	Meaning
<code>AIS_API_KEY</code>	AISStream.io key (required) — used by the monitoring areas' streams
<code>FOLLOW_AIS_API_KEY</code>	AISStream.io key for the followed ships stream. Best from a separate account (see note). Empty = reuses <code>AIS_API_KEY</code>
<code>HEATMAP_AIS_API_KEY</code>	Key of a separate AISStream account, used only for the Coverage map. Empty = feature disabled. Must be written "bare", with no comments on the same line
<code>BBOX_PRESET</code>	Area shown at startup (an area's key, e.g. <code>bari</code>)
<code>IMPORT_VF_DATA</code> / <code>IMPORT_MT_DATA</code>	<code>true</code> / <code>false</code> — enable VesselFinder / MarineTraffic import
<code>IMPORT_SF_DATA</code>	<code>true</code> / <code>false</code> — ShipFinder import + position to re-locate lost followed ships
<code>IMPORT_MST_DATA</code>	<code>true</code> / <code>false</code> — MyShipTracking import (second, independent backup position source)
<code>IMPORT_SANCTIONS</code>	<code>true</code> / <code>false</code> — screening against the OFAC SDN list
<code>IMPORT_SANCTIONS_EXTRA</code>	<code>true</code> / <code>false</code> — EU / UK OFSI / UN lists (only with <code>IMPORT_SANCTIONS</code>); default <code>true</code>
<code>IMPORT_PSC</code>	<code>true</code> / <code>false</code> — Port State Control screening (Paris/Tokyo MoU flags + banned ships)
<code>IMPORT_EQUASIS</code>	<code>true</code> / <code>false</code> — on-demand Equasis lookup in the ship detail
<code>EQUASIS_USER</code> / <code>EQUASIS_PASSWORD</code>	Equasis account credentials (free registration at equasis.org)
<code>IMPORT_GFW</code>	<code>true</code> / <code>false</code> — Global Fishing Watch enrichment; default <code>true</code>

Key	Meaning
<code>GLOBAL_FISHING_WATCH_TOKEN</code>	Global Fishing Watch API (Bearer) token. Data free for non-commercial use only
<code>ADMIN_USERNAME</code> / <code>ADMIN_EMAIL</code> / <code>ADMIN_PASSWORD</code>	Default administrator (created at startup if missing). <code>ADMIN_PASSWORD</code> is required: if it isn't configured and no admin exists yet, startup stops with an error instead of creating an account with a weak password. Change the password on a server reachable by others
<code>COOKIE_SECURE</code>	<code>true</code> / <code>false</code> — sends the session cookie only over HTTPS; set to <code>true</code> behind TLS
<code>SESSION_TTL_DAYS</code>	Login session length in days. Default <code>30</code>

💡 **Recommended: three keys from three separate AISStream accounts.** AISStream limits connections **per account, not per key**. The app opens three independent streams — **area monitoring** (`AIS_API_KEY`), **followed ships** (`FOLLOW_AIS_API_KEY`), and **coverage map** (`HEATMAP_AIS_API_KEY`). If two use keys from the **same account**, they compete for the same slot and one keeps getting rejected. Give each feature a key from a **dedicated account**.

app.config.properties — operating parameters

Tracker

Monitoraggi

Navi seguite

Mappa zone coperte

Avvia il monitoraggio

Ferma

Aree

Impostazioni

Notifiche

GNV ORION spostata da — a toscana 12:52:23 28/07/26

Area: Toscana — attiva

Manuale amministratore

Manuale utente

Log attività

494 letture

Tracking porti - Toscana

ATTIVO

Indietro

Generali

Notifiche

Aree

Integrazioni esterne

Parametri

Backup / Ripristino

Log attività

Log API

Diagnostica AIS

Le modifiche a questi parametri vengono scritte in app.config.properties e richiedono il riavvio del server per avere effetto (non basta ricaricare il browser). I segreti e le credenziali restano nel file local.properties ; i toggle di import e notifiche sono nel tab Generali.

Salva parametri 2 modifiche non salvate

RILEVAMENTO STATO NAVE

Soglia SOG (nodi) sotto cui la nave è considerata ferma 0,5 kn

Raggio (metri): ultime posizioni entro questo raggio → nave ormeggiata (isteresi anti-swing) 100 m

FINESTRE DI VISIBILITÀ

Ore: una nave in movimento viene mostrata in "Navi presenti" se vista in questa finestra 6 ore

Ore: una nave in porto (ormeggiata/all'ancora/ferma) resta in "Navi presenti" più a lungo 24 ore

CONNESSIONE E POLLING

Millisecondi: attesa prima di riconnettersi ad AISStream dopo un errore 5000 ms

Millisecondi: intervallo di aggiornamento del frontend (default 5 minuti) 300000 ms

Settings — Parameters: configuration fields grouped by category with description.

Contains thresholds and parameters (time windows, radii, retention, berths, score weights). Format KEY=value , each parameter documented by a comment in the file. **You can also edit them from the interface at ⚙ Settings → Parameters** (more convenient). Either way, you need to **restart the server** to apply them (read only at startup). Examples:

Key	Meaning	Default
<code>SOG_FERMA_KN</code>	Speed (knots) below which a ship is “stationary”	<code>0.5</code>
<code>ACTIVE_WINDOW_HOURS</code>	Hours a moving ship stays among the “current” ones	<code>6</code>
<code>PORT_WINDOW_HOURS</code>	Hours a ship in port stays among the “current” ones	<code>24</code>
<code>POLL_INTERVAL_MS</code>	Interface refresh interval (ms)	<code>300000</code>
<code>AIS_OUTAGE_CHECK</code>	Enables AIS outage detection	<code>true</code>
<code>AIS_OUTAGE_SILENCE_MIN</code>	Minutes without signals before querying the uptime monitor	<code>10</code>
<code>AIS_UPTIME_SELFHOST_URL</code>	URL of your own self-hosted uptime monitor instance (queried first)	<i>(empty)</i>
<code>AIS_UPTIME_URL</code>	URL of the public uptime monitor, used as a fallback	<code>https://aisuptime.buttermi</code>
<code>AREA_SILENT_THRESHOLD_MIN</code>	Minutes without AIS signal before an area is “in <u>fallback mode</u> ”	<code>20</code>
<code>FALLBACK_MAX_REQ_PER_HOUR</code>	Max requests/hour (ShipFinder+MyShipTracking combined, all areas) in fallback mode	<code>90</code>
<code>FALLBACK_CIRCUIT_TRIP_COUNT</code>	403/429 failures (across distinct ships) that pause a source	<code>5</code>

Key	Meaning	Default
FALLBACK_CIRCUIT_TRIP_WINDOW_MIN	Time window for counting those failures	10
FALLBACK_CIRCUIT_COOLDOWN_MIN	Minutes a source stays paused after a suspected block	30
MAX_READINGS_PER_TYPE	Maximum readings kept per message type	10000
BERTH_CLUSTER_EPS_M	Mooring-to-berth clustering radius (meters)	80
BERTH_MIN_PTS	Minimum nearby moorings to form a berth	3
BERTH_MIN_MOORINGS	Minimum moorings before characterizing a berth	10
BERTH_DOMINANT_PCT	Percentage a category must exceed to name the berth	60
BERTH_RECOMPUTE_MIN	Minutes between one automatic berth recomputation and the next	30
HEATMAP_GRID_DEG	Coverage map cell size, in degrees (~28 km)	0.25
HEATMAP_FLUSH_SEC	How often (seconds) counts are written to disk	10
GROUP_ACTIVITY_LOG_RETENTION_DAYS	Retention (days) of the group activity log (see above)	90

Key	Meaning	Default
TRANSIT_STOP_MIN_H	Minimum hours in an area for a visit to count as a call rather than a mere crossing (transit-area search and area-change notification)	3
TRANSIT_STOP_MAX_SOG_KN	Minimum observed speed below which the ship counts as genuinely stopped (recent visits only, whose positions are still stored)	0.5
TRANSIT_MIN_KN	Minimum implied average speed of a direct passage between two areas: below it, the elapsed time counts as too long	4
TRANSIT_MIN_SLACK_H	Floor of the time limit, so nearby areas are not penalised	12
TRANSIT_MAX_GAP_DAYS	Cap of the time limit, whatever the distance	30
TRANSIT_MAX_ROWS	Maximum ships returned by one transit-area search	500
AREA_CHANGE_REQUIRE_STOP	Fire the area-change notification only if the ship called at the origin area	true
AREA_CHANGE_REQUIRE_PLAUSIBLE_TIME	Fire it only if the origin call is recent enough to explain the arrival	true
AREA_CHANGE_SKIP_OVERLAPPING	No area-change notification between two areas whose boxes overlap	true

Key	Meaning	Default
<code>RISK_*</code>	Risk score weights and thresholds (see comments in the file)	various

`bounding-boxes.json` — area definitions

Lists the monitoring areas. **The recommended way to manage them is the  Areas screen** (which rewrites this file on its own). You can edit it by hand for initial provisioning; if so, **restart** the app.

Each area:

```
"bari": { "name": "Porto di Bari", "keyword": "BARI", "sw": [40.95, 16.60], "ne": [41.30, 16.60] }
```

Field	Meaning
<code>key (bari)</code>	Internal area identifier (also used by <code>BBOX_PRESET</code>)
<code>name</code>	Name shown in the interface
<code>keyword</code>	(Optional, can be <code>null</code>) filters “Expected ships” by destination
<code>sw</code>	South-West corner <code>[lat, lon]</code> in decimal degrees
<code>ne</code>	North-East corner <code>[lat, lon]</code> in decimal degrees

Manual edits to this file while the app is running only take effect after a restart. If you use the Areas screen, the file’s formatting gets normalized (stays valid, but indentation changes).

Backup, restore, and deploy

The screenshot shows the Tracker application interface. On the left is a sidebar with navigation options: Monitoraggi, Navi seguite, Mappa zone coperte, Avvia il monitoraggio, Ferma, Aree, Impostazioni, Notifiche, GNV ORION spostata da — a toscana, Area: Toscana — attiva, Manuale amministratore, Manuale utente, Log attività, and 494 letture. The main panel is titled 'Tracking porti - Toscana' and has a status 'ATTIVO'. It contains a tabbed interface with 'Backup / Ripristino' selected. The 'Backup automatico' section shows a list of automatic backups with columns for type, date, time, and size, and a 'Ripristina' button for each. The 'BACKUP COMPLETO' section has 'Esporta tutto' and 'Importa tutto' buttons. The 'DATI' section has 'Esporta CSV' and 'Backup database' buttons. The top right shows user roles: Amministratore, Admin, and Esci.

Settings — Backup tab: download and export data, restore a backup.

From **Settings** → **Backup** (visible only to administrators) you **download a backup** of the database, **restore** a saved backup, and **export** data.

- **Restoring** a database replaces **all** current data (irreversible): download a backup first. After restoring, data is reassigned to the correct area based on coordinates. Restoring **does not** re-trigger VesselFinder/MarineTraffic scraping (the data is already in the restored DB).
- **Coverage map** data lives in a **separate database**, exportable/importable on its own and still included in the full backup.
- **Auto-restore after a deploy**: the database is wiped whenever the application is updated. If, at startup, the DB **doesn't exist** and there are **auto-backups** present (`data/backups/` folder), the app automatically restores the latest backup. Requires the backups folder to survive the deploy. Doesn't trigger if the DB exists but was simply emptied with "Clear data". Can be disabled with `AUTO_RESTORE_ON_DEPLOY=false` in `app.config.properties`.

Credits

AIS outage detection (the outage banner) relies on the [AISStream-Uptime](https://github.com/buttermilkgreen/AISStream-Uptime) project by buttermilkgreen, an uptime monitor for the AISStream service. The app doesn't embed its code: it only queries the public API of its hosted instance (<https://aisuptime.buttermilkgreen.fyi>) to figure out whether a data silence is due to a service outage or simply a quiet area. The project is open source (MIT license) and you can **self-host it**: point `AIS_UPTIME_SELFHOST_URL` at your own instance's URL.